

BASEBALL AND STATISTICS

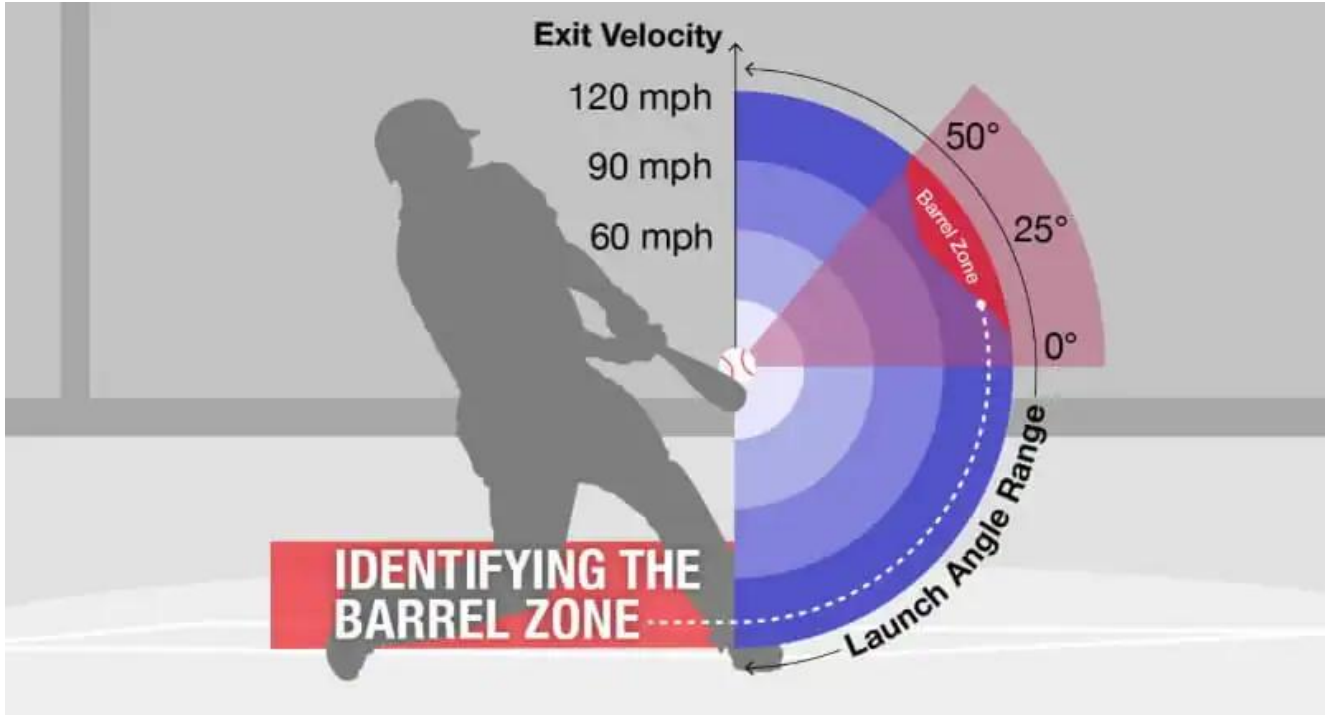
SESSION 3-IN-GAME METRICS AND STRATEGY

Review of Session 2

- Advanced Metrics
  - Hitting
  - Pitching
  - Fielding
- Player Analysis
  - Top Players of All-Time
  - HOF Minnesota Twins
  - 2026 Award Winners



In-Game Metrics



More finely grained metrics

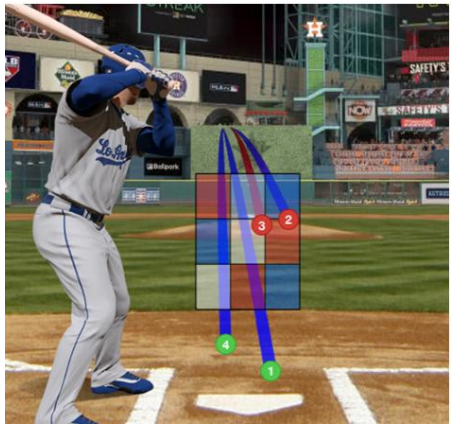
- The Moneyball Era was marked by seasonal statistics available in database form as SQL, MS Access, and R - see course website for download links
- Retrosheet game-by-game data. Retrosheet is a volunteer-run project that provides historical baseball game data, box scores, and play-by-play accounts dating back to the late 19th century. Data is available in CSV form
- In-game analytics collected live on a pitch-by-pitch basis

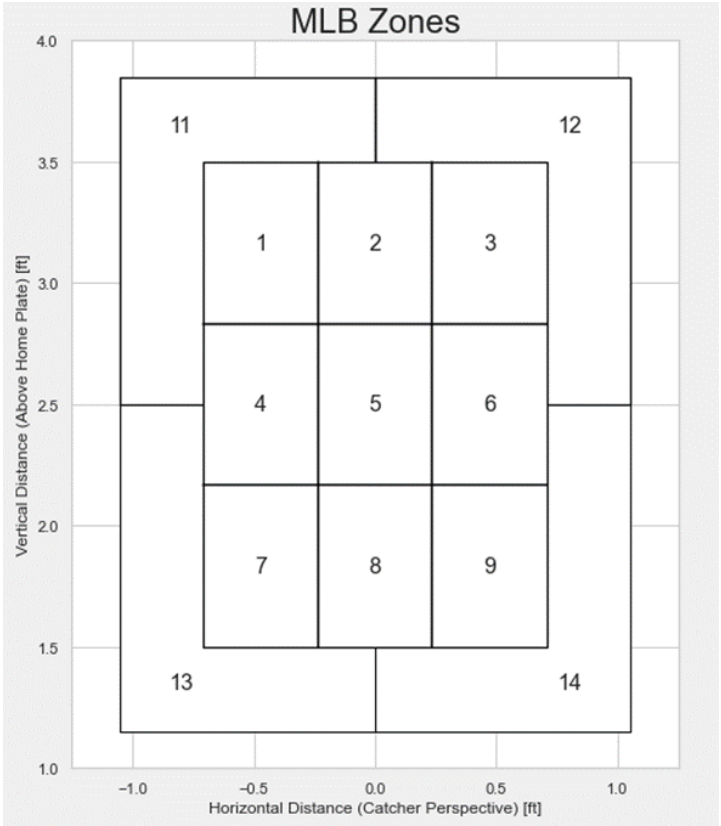
Different flavors of in-game analytics

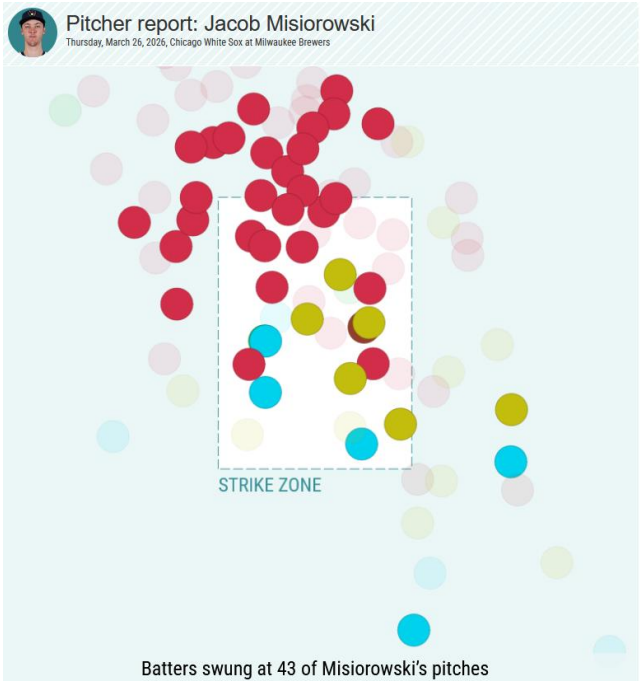
Hawk-Eye optical cameras have provided ball, player, and bat tracking data for Statcast in all 30 MLB parks since 2019, recording every pitch, hit, throw, and player.

- Some data is shown live to fans: broadcast graphics and the ABS challenge animations on ballpark scoreboards and home TV.
- Some is used by teams during games: (1) MLB-issued dugout iPads showing delayed video, (2) PitchCom for pitch signals, (3) the ABS challenge system
- Some is used for teams' pregame analysis: scouting reports, pitcher-batter matchup models, pitch plans, defensive positioning cards. The boundary between this pregame bucket and live in-game use was just redrawn by MLB this year.

Pitch Location Data Available to Fans







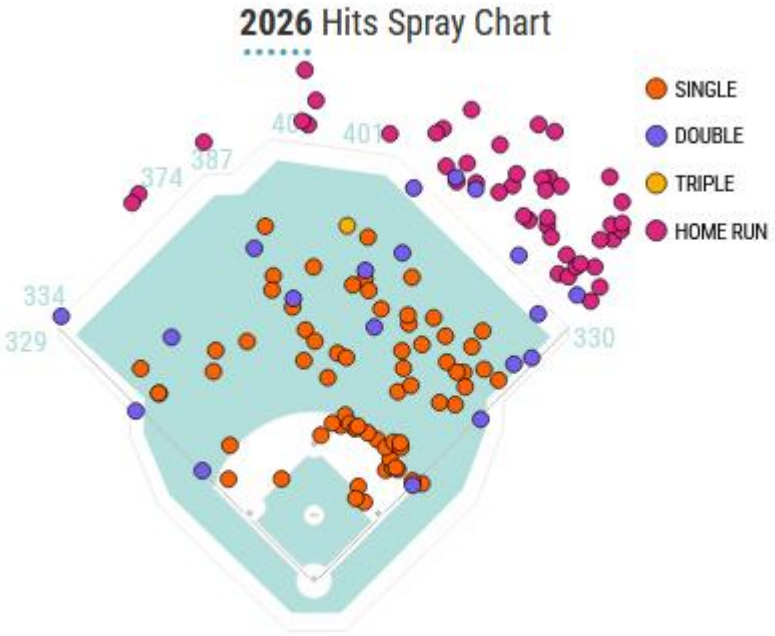
Statcast data that teams may use in-game to assist pitcher and batter substitution decisions.

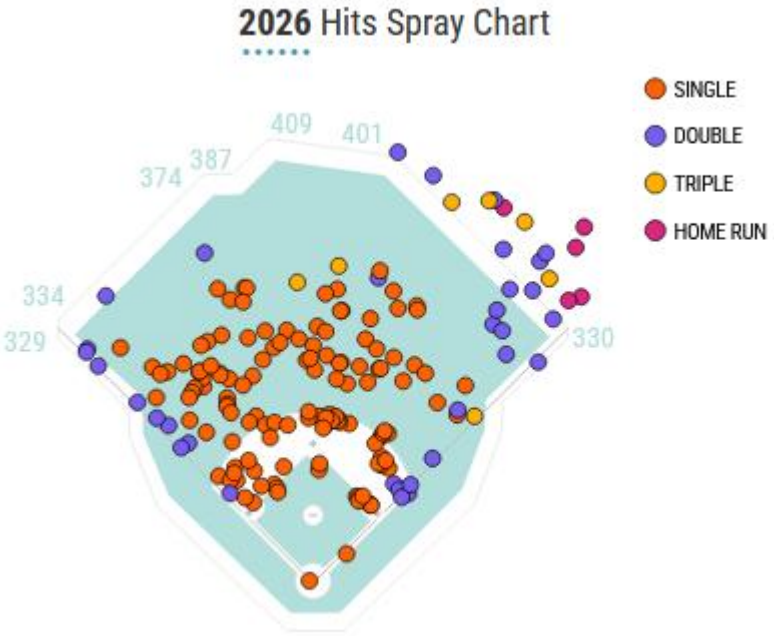
- Bullpen management and the “three times through the order rule” may be the most controversial issue. The Dodgers and the Rays illustrate very different perspectives.
- According to an article in Dodger Blue, Dave Roberts and his coaches use “third time through risk” as well as pitch shape, recovery schedules, bullpen freshness, and the next day’s needs, combined with coaches’ real-time observation of players’ body language.
- At the opposite extreme, *in 2023* the Rays and their brain trust including manager Kevin Cash leaned much more heavily into occasional use of starting pitchers for just a single inning.
- Might these two perspectives be somewhat explained by the experience of their starters?

Strategies in using PitchCom

- The catcher wears a forearm sleeve with nine buttons; the pitcher hears the call through a receiver in his cap; up to three other fielders can also receive, allowing them to adjust positioning. Three calling methods now coexist: catcher-called, pitcher-called, and dugout-called.
- In September 2025, Miami became the first MLB team to call pitches from the dugout, using a system assistant pitching coach Rob Marcello had been developing since 2019. A coach behind the dugout railing flashes numbered hand signals to the catcher, who enters the call into PitchCom. The stated goals: remove emotional bias, free young catchers to focus on framing and throwing, and apply scouting reports and real-time velocity and movement data too extensive to fit on a catcher's wristband. ESPN reported that Miami's ERA fell from 4.60 to 4.14, its home-run rate from 3.2% to 2.3%, and its two-strike strikeout rate rose from 39.7% to 42.2%. There's an AI caveat to this that we'll talk about next week.

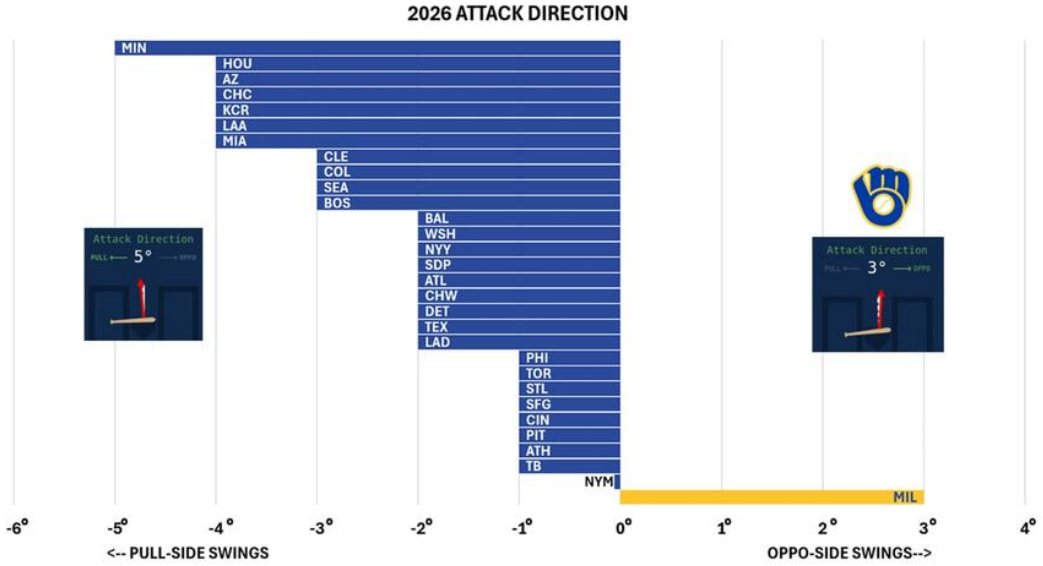
Batter Spray Charts





Luis Arraez and Kyle Schwarber – Match the spray chart to the hitter. What accounts for the difference?

Statcast data can provide fuel for discussion between Twins and Brewers fans



In 1986, Ted Williams said in an interview to Peter Gammons: "the toughest thing is to pull the ball and get it in the air ... that's the premium, par excellence hit."

[MLB.com article by Mike Petriello]

ABS visuals


ABS Strike Zone
STRIKE ZONE WIDTH
17"
Plate Width: 17"
TOP AND BOTTOM
53.5% height
27% height
STRIKE ZONE DEPTH
8.5"
8.5"


ABS STRIKE ZONE
HEIGHT PERCENTAGE
Jose Altuve: 5' 6"
Aaron Judge: 6' 7"
TOTAL HEIGHT 1' 5"
TOTAL HEIGHT 1' 9"
Top height: 53.5%
Bottom height: 27%
Total height is rounded
Source: MLB

Strategies in using ABS challenges

- Only the batter, pitcher, or catcher may challenge by tapping their cap or helmet within two seconds of the call. No help from the dugout permitted.
- Each team starts with two challenges and keeps a challenge when it succeeds.
- The verdict is computed from the same Hawk-Eye tracking that powers Statcast, using a player-specific zone: 17 inches wide, spanning 27% to 53.5% of the batter's measured height, with ball location measured on a plane 8.5 inches behind the front of the plate.
- An animated graphic of the pitch location is shown on the videoboard and broadcast before play resumes.
- Early 2026 results (first 47 games, per ESPN): 175 challenges, 94 overturned. Catchers were by far the best challengers: 59 of 92 successful (64.1%) vs hitters at 42.3%.

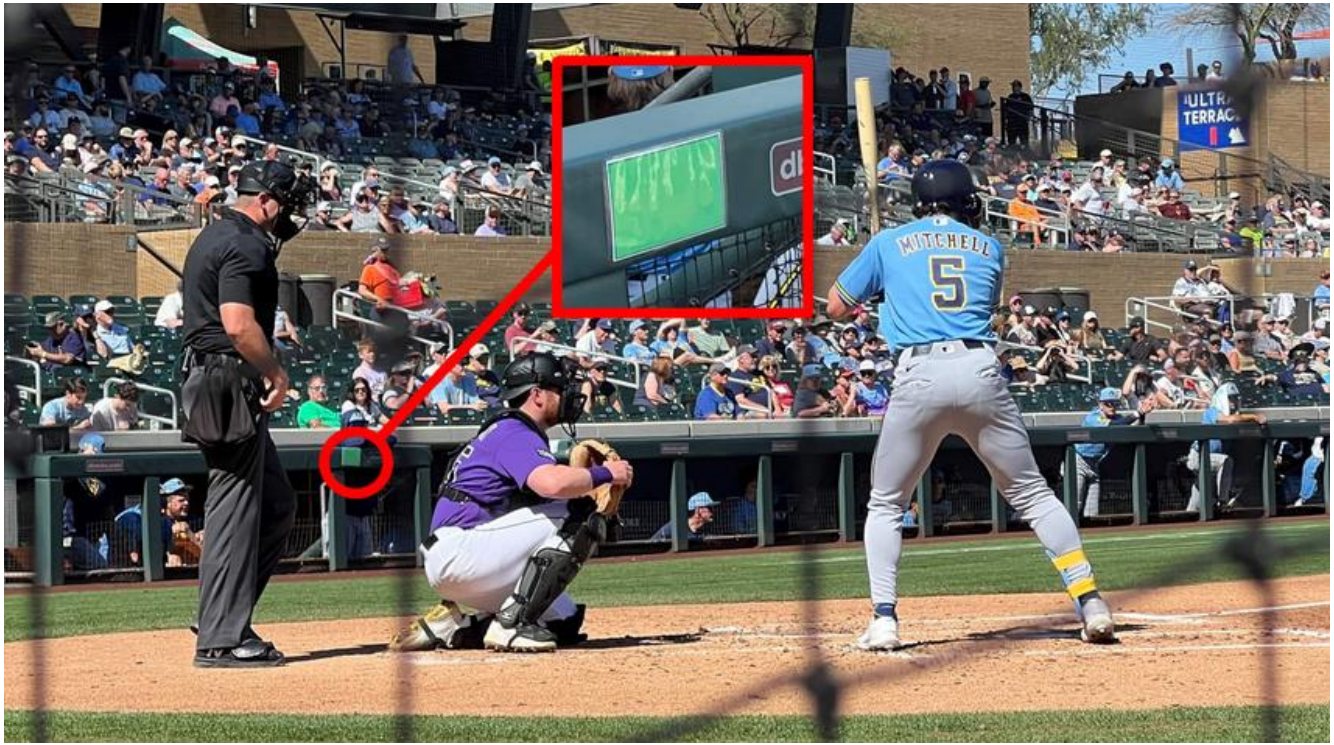
ABS visuals with the complication of a crouch



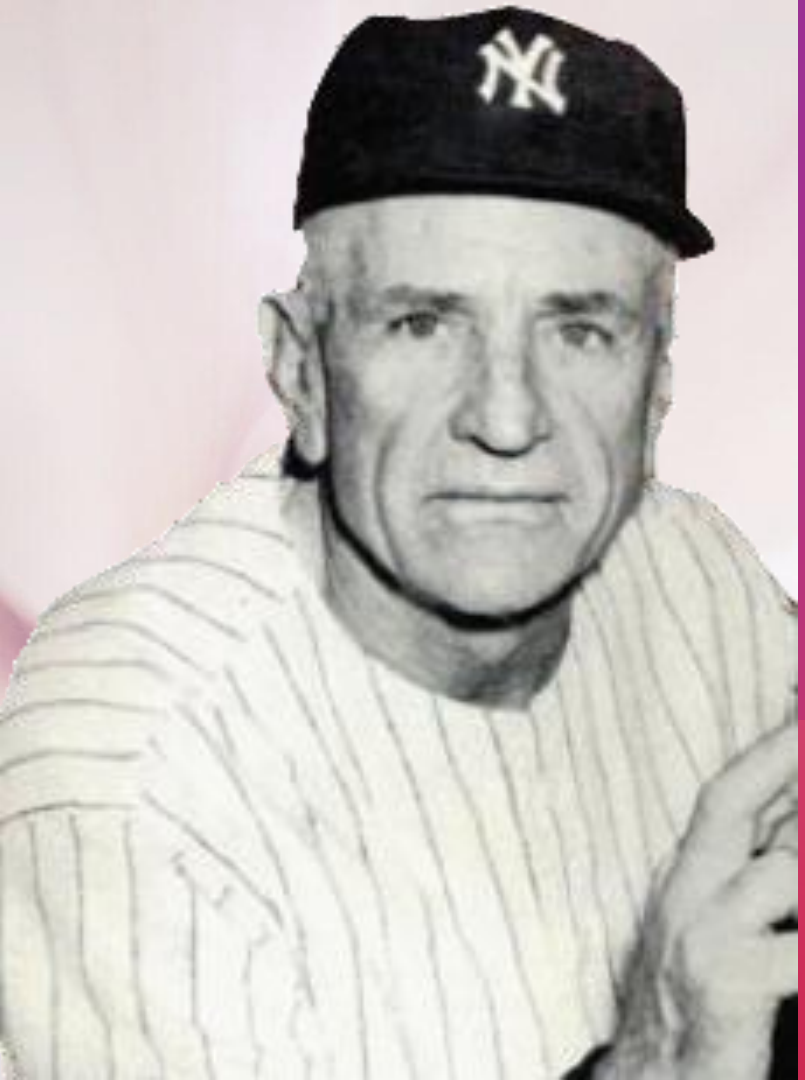


Eddie Gaedel at 3’7” ?





[How a green index card this spring honed Brewers' ABS approach ... before it couldn't]



STRATEGY

ROSTER BUILDING

ESPN Classifications

Roster Style
Payroll Power
Homegrown Percentage
Signature Signing
Best Trade They've Made
Biggest Reclamation Project Success
Historical Baggage

Los Angeles Dodgers

Roster Style: Star Power

According to Baseball Reference, five of the top 25 position players by career OPS+ played for the Dodgers this season (Ohtani, Freeman, Betts, Tucker, Smith). Also, five of the top 20 pitchers by career ERA+ (Skubal, Yamamoto, Snell, Glasnow, Ohtani).

Payroll Power: Industry Breaking

The Dodgers top baseball's revenue charts by a good bit, and the depressing thing about that advantage for the other 29 teams is that if the Dodgers wanted to spend even more, they probably could.

2026 home attendance was 4.0M (22% higher than Yankees at 3.3M).

Other “Final Four” teams - Milwaukee 2.8M, Cleveland 2.0M, Tampa Bay 1.4M

Los Angeles Dodgers

Homegrown Percentage: 34%

The Dodgers rank 18th in homegrown percentage for their 40-man roster but just five players on their 26-man postseason roster are internally developed prospects. The Dodgers' perpetually fertile farm system is used as much for depth and for trades as it is to add to the Dodgers' star count.

Signature Signing: Shohei Ohtani

Best Trade They’ve Made: Mookie Betts

From Boston in 2020 - Traded along with David Price and cash for Alex Verdugo, Connor Wong and Jeter Downs

Los Angeles Dodgers

Biggest Reclamation Project Success - Max Muncy

Designated for Assignment by Oakland in 2017; A three time All-Star with the Dodgers

Historical Baggage: Traveling light but still some luggage here.

The Dodgers have won 27 NL pennants and 9 World Series (33% Success), compared to the Yankees (41 pennants and 27 World Series (66% Success)

Cleveland Guardians

Roster Style: Guardians-specific

Cleveland's pitching pipeline has been one of the most productive for a number of years and after a slight lag in the rotation department in recent years, the Guardians' have reasserted their ability to develop hurlers.

Cleveland's offense has a heavy emphasis on making contact and moving runners around the bases, though the Guardians have been considerably less successful than the Rays and Brewers when it comes to turning that approach into consistent runs.

One thing that sets the Guardians apart is their adoration of switch-hitters.

Payroll Power: Dormant

The Guardians continually cycle through controllable players, keeping the roster young and inexpensive, with Jose Ramirez serving as the notable exception. The Guardians' Opening Day payroll was the lowest in baseball and they've generally been bottom five in the current decade despite a lot of on-field success.

Cleveland Guardians

Homegrown Percentage: 58%

The Guardians' 40-man roster is just 1% behind St. Louis for the highest figure in baseball. The Guardians scout 'em and develop 'em.

Signature Signing: Jose Ramirez

There's nothing close to a splashy free agent signee on the Guardians' roster, but the three extensions Ramirez has signed have been immeasurably important to this franchise.

Best Trade They’ve Made: Jo Adell

From Angels in 2026 - Traded for Jacob Cozart (still in Minors)

Cleveland Guardians

Biggest Reclamation Project Success - Cade Smith

An undrafted free agent pickup in 2020. To go from off the draft board to the game's best closer in six years is a nice bit of work by Smith -- and the Guardians.

Historical Baggage: A backpack with an anvil in it

It's as simple as this: Cleveland has not won the World Series since 1948, giving it the longest active title drought in the majors.

Tampa Bay Rays

Roster Style: Ahead of the curve

The Rays are always at the extreme of some kind of spectrum.

On offense, they are aggressive, make contact and mix a heavy dose of small-ball principles that is accented by the power of Junior Caminero. In run prevention, the Rays have leaned heavily on a staff that uses changeups as a knock-out pitch.

Whatever the Rays are doing, don't try to copy it because by the time you have it implemented, they will be on to the next thing.

Payroll Power: Lean and mean

The Rays invariably land as a bottom-five payroll team. Their focus is on developing and trading for controllable players. The most talented of them are either extended early or traded in advance of an expensive free agency. The roster is always lean financially and is iterated constantly.

Tampa Bay Rays

Homegrown Percentage: 18%

All but four players on the Rays' division series roster were acquired from another organization or via free agency. Their homegrown % is the lowest in baseball.

Signature Signing: Nick Martinez

It was only a one-year deal for $9 million, plus a mutual option for 2027, but the Martinez signing last offseason was just such a Rays thing to do. Martinez fit the paradigm of what kind of staff the Rays wanted to feature. The Rays threw 750 more changeups than any other team this season.

Best Trade They’ve Made: Junior Caminero

From Guardians in November 2021 for pitcher Tobias Myers. Caminero is a two time All-Star. Myers was DFA’d by the Guardians in July 2022

Tampa Bay Rays

Biggest Reclamation Project Success - Yandy Diaz

Diaz was acquired by the Rays in a three-team 2018 trade from Cleveland. His narrative was that his power was wasted because he hit the ball on the ground far too often. He still sports a groundball rate way above the league average. He's also one of the best hitters in baseball. Over eight seasons with the Rays, he has 130 OPS+ and a batting title in 2023. This season he led the American League in hits.

Historical Baggage: Nothing to lose.

The Rays have long been identified as the team no one goes to watch that plays in what looks like an upside down plaster basin and that punches above its weight competitively through the guile and cunning of its baseball operations department. But despite all of that success and cunning, the Rays have not won a World Series.

Milwaukee Brewers

Roster Style: Young and athletic

Milwaukee began to put more emphasis on athleticism and contact hitting when it came to their position players, while encouraging situational awareness all through their system. This year's club is the most advanced manifestation of that approach.

On the pitching side, the Brewers are a mix of internally developed hurlers along with a number of acquired pitchers. Milwaukee has had an inordinate amount of success in coaxing better results from such pitchers.

Payroll Power: Bottom Third

The highest the Brewers have ranked in Opening Day payroll during their current 10-year run of success is 17th, back in 2019. They've won four straight National League Central titles while ranking no higher than 20th

Milwaukee Brewers

Homegrown Percentage: 32%

The Brewers' 40-man roster is 32% homegrown, ranking 20th; the 26-man roster is at 38%, the highest of the four remaining teams. The Brewers tend to acquire external players via trades and waiver claims as opposed to free agency.

Signature Signing: prospect extensions

The signings that have been in step with the Brewers' philosophy have been extensions, especially to some of their most talented prospects. Three starters (Jackson Chourio, Cooper Pratt and Luis Lara) each signed extensions prior to reaching the majors.

Best Trade They’ve Made: Christian Yelich

From Miami in January 2018 for four players. He won NL MVP honors in his first season and finished second in the balloting the season after. The four players the Brewers sent to Miami in the deal are all out of the majors and finished with a combined minus 5.9 career bWAR.

Milwaukee Brewers

Biggest Reclamation Project Success - Jake Bauers

This area is a specialty of the Brewers, so a typical season would see at least a couple of examples. Before 2026, Bauers had a career OPS+ of 84 and was below replacement value (minus-1.2 bWAR). This season, Bauers had a 140 OPS+, put up 3.3 bWAR and led the Brewers with 26 home runs.

Historical Baggage: Overpacked

This is an esoteric way to describe the fact that the Brewers are one of five franchises never to win a title.

Minnesota Twins

Roster Style:

Payroll Power:

The Twins ranked 24th in team payroll in 2026 (or 23 or 22, depending on source)

Minnesota Twins

Homegrown Percentage: 50%

Signature Signing:

Best Trade They've Made:
Pablo Lopez - Luis Arreaz
Mick Abel - Jhoan Duran
Taj Bradley - Griffin Jax
Joe Ryan - Nelson Cruz

Minnesota Twins

Biggest Reclamation Project Success:

Kody Clemens was DFA'd by Phillies; led 2026 Twins with 27 HRs

Historical Baggage:

The Twins last won the World Series in 1991

Longer droughts - Guardians, Padres, Brewers, Mariners, Pirates, Orioles, Tigers, Mets, Athletics, Reds

Twins lost 18 straight postseason games from 2004-2020